

AZIZ SHAMEEM

 [azizshameem.github.io](https://github.com/azizshameem) |  [Aziz-Shameem](#) |  [aziz-shameem](#)

EDUCATION

Indian Institute of Technology Bombay

B.Tech in Electrical Engineering and M.Tech in Data Science (**Cumulative GPA: 9.72/10.0**)

Mumbai, India

Nov'20 – Aug'25

SCHOLASTIC ACHIEVEMENTS

- Recipient of the **Institute Silver Medal** from IIT Bombay, for outstanding academic performance overall
- Ranked **1st** in **Electrical Engineering Dept.** and **Centre for Machine Intelligence and Data Science** (2022-25)
- Recipient of the **Summer@EPFL Travel Scholarship** (awarded to 1.6% of the applicants) (2024)
- Awarded **Institute Academic Prize** for exceptional performance in academics for **two consecutive years** (2022, 2023)
- Secured an **AIR of 62** in JEE Mains (Engineering) and an **AIR of 665** in JEE Advanced (2020)
- Conferred with **AP grade (Advanced Performer)** in **Advanced ML, Linear Algebra, Electronic Devices**

PUBLICATIONS

- Bhavya Kohli, **Aziz Shameem**, Soutrik Sarangi, Abir De, “**Learning Right Monotone Permutation Matrices for Neural Subsequence Search**”, AISTATS 2026
- M. Shukla, **A. Shameem**, M. Salzmann, A. Alahi, “**Towards Self-Supervised Covariance Estimation in Deep Heteroscedastic Regression**”, ICLR 2025
- V. Prasad, **A. Shameem**, C. White, S. Nayak, P. Jain, P. Garg, G. Ramakrishnan, “**Speeding up NAS with Adaptive Subset Selection**”, AutoML 2024

RESEARCH EXPERIENCE

Variational Auto-Encoders and Diffusion Models

May'24 – Dec'24

Summer Internship, Guide: [Prof. Alexandre alahi](#), [Prof. Amit Sethi](#)

EPFL, EE IITB

- Performed extensive analysis of the **similarities and differences** between two types of Generative Frameworks - **Variational AutoEncoders** and **Diffusion Models**
- Designed experiments to elucidate the **limitations** of VAEs, and suggested **improvements/methods** to overcome them
- Demonstrated methods for better generations by VAEs on **synthetic** as well as **benchmark datasets**
- Incorporated the methods for improving **conditional image generation using VAEs**
- Introduced **Probabilistic Optimizations** to better the computational complexity of the algorithms

OPAS : Ordered Permutation-based Alignment for Sequence matching

Jun'23 – Oct'24

Guide: [Prof. Abir De](#)

CSE, CMInDS, IIT Bombay

- Developed a **theoretical framework** based on **lagrangians** for **fast, approximate, constrained sequence matching**
- Tested the developed framework **computationally**, and obtained results on datasets of several **modalities**
- Compared the developed method with several baselines, and demonstrated over **58x speedup** over them
- Compared several **implementation variants, architectures and models** of the proposed system as ablations

PROFESSIONAL EXPERIENCE

Data Scientist

Jul'25 – Present

MEESHO Limited

Bengaluru, India

- Working on **Personalized Ranking systems** responsible for the product feeds on search and products-like-this pages
- Worked on **production analytics** and ML systems for pricing and discounting, contributing to **model development, evaluation, and iterative improvement**
- Applied statistical and algorithmic methods to support controlled experimentation and decision-making under uncertainty which included **robust reward formulations, noise-reduction strategies** and **simulation frameworks**
- Supported deployment and monitoring of models in production, collaborating with engineering teams to ensure **stability, scalability, and reliable performance**

Modem Systems Engineering Intern

May'23 – Jul'23

Qualcomm Technologies Inc.

Bengaluru, India

- Undertook extensive research on enhancing **information-deficit channel estimation** methods for **5G**
- Implemented existing as well as proposed technology in a **python simulator** and verified theoretical results
- Implemented specific **channel models** in the python simulator, to be used for future research and development

TECHNICAL SKILLS

Data Analysis/Visualization tools : Pandas, Numpy, Matplotlib, seaborn, plotly, MATLAB, MySQL

Libraries: Pytorch, Tensorflow, Numpy, Pandas, OpenCV, SciPy, Seaborn, Scikit-Learn, OpenCV, Matplotlib

Software: Git, ~~La~~TeX, MATLAB, GNU Radio, Quartus, NGSpike, Kicad, Arduino, Pycharm, Anaconda, Unity

KEY TECHNICAL PROJECTS

Implementation of Transformer Concepts in vanilla python

Apr'26

Self Project

- Extended a **minimal GPT implementation** into a modular framework by implementing modern transformer techniques including **RoPE, ALiBi, T5 Relative Position Bias, Flash Attention, Mixture-of-Experts, Multi-Token Prediction, and Byte Pair Encoding**
- Designed an extensible **model registry** and **training pipeline** for rapid experimentation with **alternative attention mechanisms, positional encodings, and language modeling architectures**
- All the implementations were done in **Vanilla Python**, with the goal of understanding tech advancements at their core by removing all the optimization black-boxes

LLM Debates - Platform for top LLMs to interact

Mar'26

Self Project

- Developed a **web platform** for Large Language Models to debate on an input prompt with configurable debate formats and automated judging
- Designed a **modular evaluation pipeline** to compare reasoning quality across models, with **experiment tracking** and an **interactive visualization interface**
- Goal was to elucidate **differences in personalities** between the top LLMs, and see if **mutual thought transfer** can benefit decision making ability

Auto-Grad Engine and GPT development

May'23 – Jul'23

Self Project

IIT Bombay

- Built an **Auto-Grad Engine** having **pytorch-like API** from scratch and demonstrated its working on a test bed
- Performed extensive study on **innate problems** in neural networks pertaining to **gradients and dead neurons**
- Implemented initialization and normalization based methods to bypass problems and demonstrated better results
- Developed a working **GPT-like architecture** from scratch and generated recognizable **Shakespearean text**

Toxicity Detection in Large Language Models

Apr'24 – May'24

Guide: [Prof. Pushpak Bhattacharya](#) (CS772 : DL for NLP)

IIT Bombay

- Trained a **text toxicity classifier** using **Recurrent Neural Nets, LSTMs** and **Transformers**
- Engineered additional features** from the analysis of textual data in an effort to improve classification of toxicity in texts
- Utilised the trained classifier network as a **filter** to mitigate toxicity in the output of **Large Language Models**

Correlated Multi-Armed Bandits

Mar'23 – Apr'23

Guide: [Prof. Jayakrishnan Nair](#), [Prof. D. Manjunath](#) (EE6106: Online Learning and Optimization)

IIT Bombay

- Performed an extensive literature review on the formulation of the **correlated multi-armed Bandit** problem
- Applied an existing framework for the **exploitation of correlation** between bandits, on different algorithms and demonstrated **substantial improvements** in the same

POSITIONS OF RESPONSIBILITY

Machine Learning Engineer

[Data Analysis and Visualisation Team](#), IITB

Jun'22 – Apr'23

- Led a team to **successful completion and publication** (On the official public handle) of analysis on **Indian Establishments and Settlements, Grading Statistics** and **Semester Exchange statistics**
- Implemented the **YOLO version-1** framework from scratch as part of a study project and gave a presentation on the same
- Delivered a comprehensive presentation on **Loss Functions for Image Restoration** and got recognized for the same

Mentor - Introduction to Machine Intelligence

[Seasons of Code](#), IITB

Apr'22 – Jul'22

- Prepared and distributed **comprehensive resources** to help beginners understand and appreciate concepts in ML
- Assisted beginners in building **AI systems** using **Reinforcement Learning** in an effort to generate interest in the field

Web Developer/Coordinator

[Techfest](#) - Asia's largest science and technology festival

Oct'21 – Feb'22

- Contributed to the creation of **TechFest Website** by developing **hover animations, Bootstrap cards and Carousels**
- Improvised and optimised existing code-base by appropriate fixes and restructuring